SIM1003 Calculus II
Tutorial 1

All even-numbered questions in the following exercises.

EXERCISE 7.6 Inverse Trigonometric Functions

Values of Inverse Trigonometric

Find the values in Exercises 2 - 8.

2. a. $\arctan(-1)$ **b.** $\tan^{-1}\sqrt{3}$ **c.** $\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

3. a. $\sin^{-1}\left(\frac{-1}{2}\right)$ **b.** $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ **c.** $\arcsin\left(\frac{-\sqrt{3}}{2}\right)$

4. a. $\sin^{-1}\left(\frac{1}{2}\right)$ **b.** $\arcsin\left(\frac{-1}{\sqrt{2}}\right)$ **c.** $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

5. a. $\cos^{-1}\left(\frac{1}{2}\right)$ **b.** $\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$ **c.** $\arccos\left(\frac{\sqrt{3}}{2}\right)$

6. a. $\csc^{-1}\sqrt{2}$ **b.** $\operatorname{arccsc}\left(\frac{-2}{\sqrt{3}}\right)$ **c.** $\csc^{-1}2$

7. a. $\operatorname{arcsec}\left(-\sqrt{2}\right)$ **b.** $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ **c.** $\sec^{-1}(-2)$

8. a. $\cot^{-1}(-1)$ **b.** $\operatorname{arccot}\left(\sqrt{3}\right)$ **c.** $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

Evaluations

Find the values in Exercises 9–12.

9. $\sin\left(\cos^{-1}\left(\frac{\sqrt{2}}{2}\right)\right)$ **10.** $\sec\left(\arccos\frac{1}{2}\right)$

11. $\tan\left(\arcsin\left(-\frac{1}{2}\right)\right)$ **12.** $\cot\left(\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$

Limits

Find the limits in Exercises 13–20. (If in doubt, look at the function's graph.)

13. $\lim_{x \to 1^-} \sin^{-1} x$

14. $\lim_{x \to -1^+} \cos^{-1} x$

15. $\lim_{x \to \infty} \tan^{-1} x$

16. $\lim_{x \to -\infty} \tan^{-1} x$

17. $\lim_{x \to \infty} \sec^{-1} x$

18. $\lim_{x \to -\infty} \sec^{-1} x$

19. $\lim_{x \to \infty} \csc^{-1} x$

20. $\lim_{x \to -\infty} \csc^{-1} x$

Finding Derivatives

In Exercises 21–42, find the derivative of y with respect to the appropriate variable.

21. $y = \cos^{-1}(x^2)$

22. $y = \cos^{-1}(1/x)$

23. $y = \arcsin \sqrt{2}\, t$

24. $y = \sin^{-1}(1 - t)$

25. $y = \sec^{-1}(2s + 1)$

26. $y = \sec^{-1} 5s$

27. $y = \csc^{-1}(x^2 + 1),\ x > 0$

28. $y = \csc^{-1} \dfrac{x}{2}$

29. $y = \sec^{-1} \dfrac{1}{t},\ 0 < t < 1$

30. $y = \arcsin \dfrac{3}{t^2}$

31. $y = \operatorname{arccot} \sqrt{t}$

32. $y = \cot^{-1} \sqrt{t - 1}$

33. $y = \ln(\tan^{-1} x)$

34. $y = \tan^{-1}(\ln x)$

35. $y = \csc^{-1}(e^t)$

36. $y = \arccos(e^{-t})$

37. $y = s\sqrt{1 - s^2} + \cos^{-1} s$

38. $y = \sqrt{s^2 - 1} - \sec^{-1} s$

39. $y = \tan^{-1} \sqrt{x^2 - 1} + \csc^{-1} x,\quad x > 1$

40. $y = \cot^{-1} \dfrac{1}{x} - \tan^{-1} x$

41. $y = x \arcsin x + \sqrt{1 - x^2}$

42. $y = \ln(x^2 + 4) - x \tan^{-1}\left(\dfrac{x}{2}\right)$

Evaluating Integrals

Evaluate the integrals in Exercises 47–70.

47. $\displaystyle\int \frac{dx}{\sqrt{9 - x^2}}$

48. $\displaystyle\int \frac{dx}{\sqrt{1 - 4x^2}}$

49. $\displaystyle\int \frac{dx}{17 + x^2}$

50. $\displaystyle\int \frac{dx}{9 + 3x^2}$

51. $\displaystyle\int \frac{dx}{x\sqrt{25x^2 - 2}}$

52. $\displaystyle\int \frac{dx}{x\sqrt{5x^2 - 4}}$

53. $\displaystyle\int_0^1 \frac{4\,ds}{\sqrt{4 - s^2}}$

54. $\displaystyle\int_0^{3\sqrt{2/4}} \frac{ds}{\sqrt{9 - 4s^2}}$

55. $\displaystyle\int_0^2 \frac{dt}{8 + 2t^2}$

56. $\displaystyle\int_{-2}^2 \frac{dt}{4 + 3t^2}$

57. $\displaystyle\int_{-1}^{-\sqrt{2/2}} \frac{dy}{y\sqrt{4y^2 - 1}}$

58. $\displaystyle\int_{-2/3}^{-\sqrt{2/3}} \frac{dy}{y\sqrt{9y^2 - 1}}$

59. $\displaystyle\int \frac{3\,dr}{\sqrt{1 - 4(r - 1)^2}}$

60. $\displaystyle\int \frac{6\,dr}{\sqrt{4 - (r + 1)^2}}$

61. $\displaystyle\int \frac{dx}{2 + (x - 1)^2}$

62. $\displaystyle\int \frac{dx}{1 + (3x + 1)^2}$

63. $\displaystyle\int \frac{dx}{(2x - 1)\sqrt{(2x - 1)^2 - 4}}$

64. $\displaystyle\int \frac{dx}{(x + 3)\sqrt{(x + 3)^2 - 25}}$

EXERCISE 7.7 Hyperbolic Functions

Rewrite the expressions in Exercises 5–10 in terms of exponentials and simplify the results as much as you can.

5. $2 \cosh (\ln x)$

6. $\sinh (2 \ln x)$

7. $\cosh 5x + \sinh 5x$

8. $\cosh 3x - \sinh 3x$

9. $(\sinh x + \cosh x)^4$

10. $\ln (\cosh x + \sinh x) + \ln (\cosh x - \sinh x)$

11. Prove the identities

$$\sinh (x + y) = \sinh x \cosh y + \cosh x \sinh y,$$
$$\cosh (x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

Then use them to show that

a. $\sinh 2x = 2 \sinh x \cosh x.$

b. $\cosh 2x = \cosh^2 x + \sinh^2 x.$

12. Use the definitions of $\cosh x$ and $\sinh x$ to show that

$$\cosh^2 x - \sinh^2 x = 1.$$

Finding Derivatives

In Exercises 13–24, find the derivative of y with respect to the appropriate variable.

13. $y = 6 \sinh \frac{x}{3}$

14. $y = \frac{1}{2} \sinh (2x + 1)$

15. $y = 2\sqrt{t} \tanh \sqrt{t}$

16. $y = t^2 \tanh \frac{1}{t}$

17. $y = \ln (\sinh z)$

18. $y = \ln (\cosh z)$

19. $y = (\operatorname{sech} \theta)(1 - \ln \operatorname{sech} \theta)$

20. $y = (\operatorname{csch} \theta)(1 - \ln \operatorname{csch} \theta)$

21. $y = \ln \cosh v - \frac{1}{2} \tanh^2 v$

22. $y = \ln \sinh v - \frac{1}{2} \coth^2 v$

23. $y = (x^2 + 1) \operatorname{sech} (\ln x)$

(*Hint:* Before differentiating, express in terms of exponentials and simplify.)

24. $y = (4x^2 - 1) \operatorname{csch} (\ln 2x)$

In Exercises 25–36, find the derivative of y with respect to the appropriate variable.

25. $y = \sinh^{-1}\sqrt{x}$

26. $y = \cosh^{-1} 2\sqrt{x+1}$

27. $y = (1-\theta)\tanh^{-1}\theta$

28. $y = (\theta^2 + 2\theta)\tanh^{-1}(\theta+1)$

29. $y = (1-t)\coth^{-1}\sqrt{t}$

30. $y = (1-t^2)\coth^{-1} t$

31. $y = \cos^{-1} x - x\,\text{sech}^{-1} x$

32. $y = \ln x + \sqrt{1-x^2}\,\text{sech}^{-1} x$

33. $y = \text{csch}^{-1}\left(\frac{1}{2}\right)^{\theta}$

34. $y = \text{csch}^{-1} 2^{\theta}$

35. $y = \sinh^{-1}(\tan x)$

36. $y = \cosh^{-1}(\sec x), \quad 0 < x < \pi/2$

EXERCISE 8.1 Using Basic Integration Formulas

Assorted Integrations

The integrals in Exercises 1–44 are in no particular order. Evaluate each integral using any algebraic method or trigonometric identity you think is appropriate. When necessary, use a substitution to reduce it to a standard form.

1. $\displaystyle\int_0^1 \frac{16x}{8x^2+2}\,dx$

2. $\displaystyle\int \frac{x^2}{x^2+1}\,dx$

3. $\displaystyle\int (\sec x - \tan x)^2\,dx$

4. $\displaystyle\int_{\pi/4}^{\pi/3} \frac{dx}{\cos^2 x \tan x}$

5. $\displaystyle\int \frac{1-x}{\sqrt{1-x^2}}\,dx$

6. $\displaystyle\int \frac{dx}{x-\sqrt{x}}$

7. $\displaystyle\int \frac{e^{-\cot z}}{\sin^2 z}\,dz$

8. $\displaystyle\int \frac{2^{\ln z^3}}{16z}\,dz$

9. $\displaystyle\int \frac{dz}{e^z + e^{-z}}$

10. $\displaystyle\int_1^2 \frac{8\,dx}{x^2-2x+2}$

11. $\displaystyle\int_{-1}^{0} \frac{4\,dx}{1+(2x+1)^2}$

12. $\displaystyle\int_{-1}^{3} \frac{4x^2-7}{2x+3}\,dx$

13. $\displaystyle\int \frac{dt}{1-\sec t}$

14. $\displaystyle\int \csc t \sin 3t\,dt$

15. $\displaystyle\int_0^{\pi/4} \frac{1+\sin\theta}{\cos^2\theta}\,d\theta$

16. $\displaystyle\int \frac{d\theta}{\sqrt{2\theta-\theta^2}}$

17. $\displaystyle\int \frac{\ln y}{y + 4y \ln^2 y}\, dy$

18. $\displaystyle\int \frac{2^{\sqrt{y}}\, dy}{2\sqrt{y}}$

19. $\displaystyle\int \frac{d\theta}{\sec\theta + \tan\theta}$

20. $\displaystyle\int \frac{dt}{t\sqrt{3 + t^2}}$

21. $\displaystyle\int \frac{4t^3 - t^2 + 16t}{t^2 + 4}\, dt$

22. $\displaystyle\int \frac{x + 2\sqrt{x - 1}}{2x\sqrt{x - 1}}\, dx$

23. $\displaystyle\int_0^{\pi/2} \sqrt{1 - \cos\theta}\, d\theta$

24. $\displaystyle\int (\sec t + \cot t)^2\, dt$

25. $\displaystyle\int \frac{dy}{\sqrt{e^{2y} - 1}}$

26. $\displaystyle\int \frac{6\, dy}{\sqrt{y}\,(1 + y)}$

27. $\displaystyle\int \frac{2\, dx}{x\sqrt{1 - 4\ln^2 x}}$

28. $\displaystyle\int \frac{dx}{(x - 2)\sqrt{x^2 - 4x + 3}}$

29. $\displaystyle\int (\csc x - \sec x)(\sin x + \cos x)\, dx$

30. $\displaystyle\int 3 \sinh\left(\frac{x}{2} + \ln 5\right) dx$

31. $\displaystyle\int_{\sqrt{2}}^{3} \frac{2x^3}{x^2 - 1}\, dx$

32. $\displaystyle\int_{-1}^{1} \sqrt{1 + x^2}\, \sin x\, dx$

33. $\displaystyle\int_{-1}^{0} \sqrt{\frac{1 + y}{1 - y}}\, dy$

34. $\displaystyle\int e^{z + e^z}\, dz$

35. $\displaystyle\int \frac{7\, dx}{(x - 1)\sqrt{x^2 - 2x - 48}}$

36. $\displaystyle\int \frac{dx}{(2x + 1)\sqrt{4x + 4x^2}}$

37. $\displaystyle\int \frac{2\theta^3 - 7\theta^2 + 7\theta}{2\theta - 5}\, d\theta$

38. $\displaystyle\int \frac{d\theta}{\cos\theta - 1}$

39. $\displaystyle\int \frac{dx}{1 + e^x}$

Hint: Use long division.

40. $\displaystyle\int \frac{\sqrt{x}}{1 + x^3}\, dx$

Hint: Let $u = x^{3/2}$.

41. $\displaystyle\int \frac{e^{3x}}{e^x + 1}\, dx$

42. $\displaystyle\int \frac{2^x - 1}{3^x}\, dx$

43. $\displaystyle\int \frac{1}{\sqrt{x}\,(1 + x)}\, dx$

44. $\displaystyle\int \frac{\tan\theta + 3}{\sin\theta}\, d\theta$